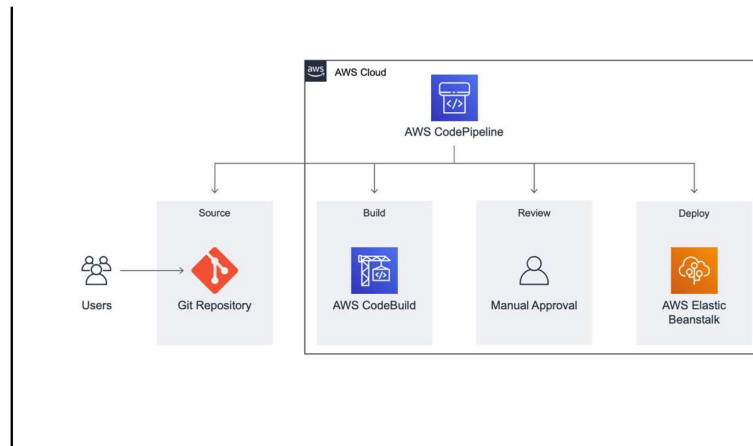


Name : Vivek Chaudhari

Project: 11

DeploySync : Automated attendance app deployment pipeline



Step 1: Create Project

- Built attendance app (frontend + backend)
- Used HTML, CSS, JS (frontend)
- Used Flask (backend)

◆ Step 2: Upload to GitHub

- Created repository
- Uploaded full project structure

◆ Step 3: Setup S3 (Frontend Hosting)

- Created S3 bucket
- Enabled static website hosting
- Uploaded frontend files
- Made bucket public

◆ Step 4: Setup EC2 (Backend)

- Launched Ubuntu EC2 instance
 - Installed Python + Flask
 - Ran backend on port 5000
 - Opened port 5000 in security group
-

◆ **Step 5: Setup CI/CD Pipeline**

- Created pipeline using **AWS CodePipeline**
 - Connected GitHub repository
 - Added build stage using CodeBuild
 - Deployed frontend to S3 automatically
-

◆ **Step 6: Backend Deployment (Manual Script)**

- Created deploy.sh script
 - Script:
 - Pull latest code
 - Restart backend server
 - Executed manually on EC2
-

◆ **Step 7: Testing**

- Made changes in frontend
 - Pushed to GitHub
 - Verified auto deployment on S3
 - Tested backend via EC2 public IP
-

Final Output

- ✓ Frontend auto deploy (S3)
- ✓ Backend running on EC2

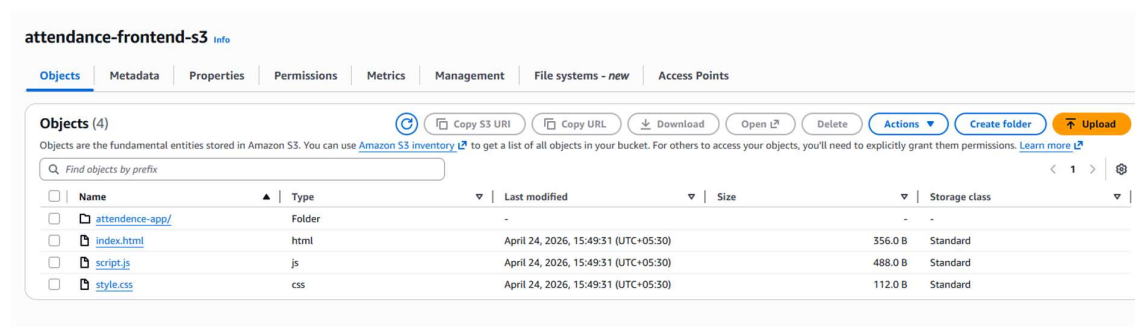
- ✓ CI/CD pipeline working
- ✓ Real-time updates on push

🧠 Final Summary (For Report)

This project implements a CI/CD pipeline for an attendance application using AWS services. The frontend is hosted on Amazon S3 and automatically deployed using AWS CodePipeline and CodeBuild whenever changes are pushed to GitHub. The backend runs on an EC2 instance and is updated using a deployment script. This setup reduces manual effort and ensures faster and reliable application updates.

SnapShot:

AWS S3 Bucket



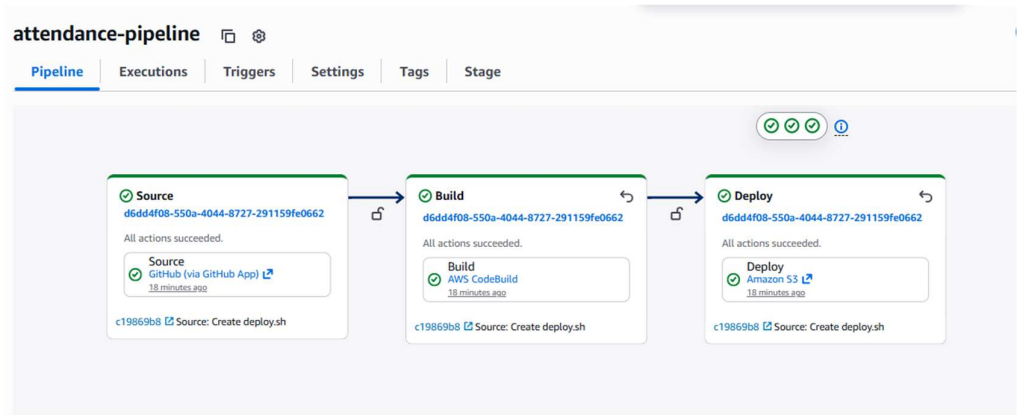
attendance-frontend-s3 Info						
Objects	Metadata	Properties	Permissions	Metrics	Management	File systems - new
Access Points						
Objects (4)						
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more						
Find objects by prefix						
☐	Name	Type	Last modified	Size	Storage class	
☐	attendance-app/	Folder	-	-	-	
☐	index.html	html	April 24, 2026, 15:49:31 (UTC+05:30)	356.0 B	Standard	
☐	script.js	js	April 24, 2026, 15:49:31 (UTC+05:30)	488.0 B	Standard	
☐	style.css	css	April 24, 2026, 15:49:31 (UTC+05:30)	112.0 B	Standard	

Final Browser Output with Backend

<http://attendance-frontend-s3.s3-website.ap-south-1.amazonaws.com>

Attendance System

AWS Code Pipeline Create for Auto Deployment



AWS EC2 for Backend

Instance summary for i-091d4fceebea29802b (attendance-backend) Info		
Updated 1 minute ago		
Instance ID i-091d4fceebea29802b	Public IPv4 address 13.203.78.20 open address	Private IPv4 addresses 172.31.14.174
IPv6 address -	Instance state Running	Public DNS ec2-13-203-78-20.ap-south-1.compute.amazonaws.com open a...
Hostname type IP name: ip-172-31-14-174.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-14-174.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn m...
Auto-assigned IP address 13.203.78.20 [Public IP]	VPC ID vpc-07abdf21c0b27f1179	Auto Scaling Group name -
IAM role -	Subnet ID subnet-09a155f615754ca22	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:216153283160:instance/i-091d4fceebea29802b	
Operator		